NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Introduction to Internet of Things

Assignment-Week 1

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15** **Total marks: 15 X 1= 15**

QUESTION 1:

Which of the following areas are identified as IoT enablers?

a. RFID
b. Smart Networks
c. Nanotechnology
d. All of the above

Correct Answer: d. All of the above

Detailed Solution: IoT enablers connect devices and systems using several technologies and protocols to the Internet for different uses.
See lecture 1 @ 14:45

QUESTION 2:

Which of the following is a digital sensor?

a. KY-013
b. LX1972
c. DHT11
d. LM35D

Correct Answer: c. DHT11

Detailed Solution: LX1972 is an analog light sensor. KY-013 and LM35D are analog temperature sensors. DHT11 is a basic, low-cost digital temperature and humidity sensor.

See lecture 3 (Sensing)

QUESTION 3:

Which of the following is considered as long range radio used in IoT?

a. Bluetooth
b. RFID
c. LoRa
d. None of them

Correct Answer: c. LoRa

Detailed Solution: Bluetooth and RFID are usually considered short range solutions with signal range typically ranging up to 50m maximum. LoRa stands for Long Range, is a spread spectrum modulation technique with range from 2-5 km (urban), 5-15 km (rural), and >15 km (LOS).

QUESTION 4:

Solenoid valve finds application in _____ equipment.

a. Both hydraulic & electrical
b. Both electrical & pneumatic
c. Both electrical & mechanical
d. Both hydraulic & pneumatic

Correct Answer: d. Both hydraulic & pneumatic

Detailed Solution: A solenoid valve is an electrically activated valve, typically used to control the flow or direction of air or liquid in fluid power systems. It finds applications in both pneumatic and hydraulic fluid power systems.
See lecture 4 @ 5:40

QUESTION 5:

Which of the following connects an IoT LAN with a WAN, and to the Internet?

a. IoT node
b. IoT proxy
c. IoT gateway
d. IoT Network

Correct Answer: c. IoT gateway

Detailed Solution: IoT gateway is a router connecting an IoT LAN with a WAN, and to the Internet. An IoT gateway is a layer 3 device, which forwards IP packets between the IoT LAN and WAN implementing both the network technologies.
See lecture 2 (IPv6 Addressing Strategies).

QUESTION 6:

Which of the following is NOT a function of IoT Gateways?

a. Switching
b. Firewall & VPN services
c. Protocol conversion
d. None of the above

Correct Answer: d. None of the above

Detailed Solution: An IoT gateway performs several critical functions, like switching, routing, translating protocols, firewall &VPN services, security, processing, managing and filtering data.
See lecture 5 @ 20:20

QUESTION 7:

Which of the following is NOT a characteristic of IoT?

a. Intermittent connectivity
b. Ambiguous naming & addressing
c. Scalable architecture
d. Abundance of sleeping nodes

Correct Answer: b. Ambiguous naming & addressing
Detailed Solution: Efficient, scalable, and associated architecture; unambiguous naming and addressing; abundance of sleeping nodes, mobile and non-IP devices; and intermittent connectivity are few of the characteristics of IoT.
See lecture 1 @ 16:05

QUESTION 8:

Rack & pinion is an example of _____ actuator.

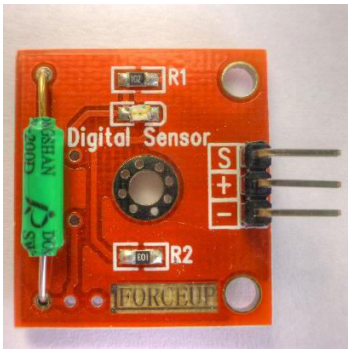
a. Hydrodynamic
b. Electrical
c. Mechanical
d. None of these

Correct Answer: c. Mechanical

Detailed Solution: A rack and pinion is a type of linear actuator that comprises a circular gear (the pinion) engaging a linear gear (the rack), which convert rotational motion into linear motion.
See lecture 4 @ 14:20

QUESTION 9:

Identify the component shown below.



a. Gas sensor
b. Tilt sensor
c. Light sensor
d. Speaker

Correct Answer: b. Tilt sensor

Detailed Solution: Tilt sensors measure the tilting position with reference to gravity and are used in numerous applications. They enable the easy detection of orientation or inclination.
See lecture 3 (Sensing)

QUESTION 10:

Which of the following is NOT true?

a. M2M is not a subset of IoT
b. IoT is a subset of CPS
c. IoT and M2M are same
d. All of the above

Correct Answer: d. All of the above

Detailed Solution: M2M is a part of the IoT, where IoT has a broader scope than M2M. Both are different since M2M uses point-to-point communication. IoT is a subset of CPS.
See lecture 1 @ 33:50

QUESTION 11:

"IoT gateway WAN address changes without change in LAN address" - Which of the following helps in achieving it?

a. Unique local address
b. Network Prefix Translation
c. Tunneling

d. None of the above

Correct Answer: a. Unique local address

Detailed Solution: ULA are designed to provide adequate uniqueness in order to have extremely small risk of address collision.
See lecture 2 @ 9:25

QUESTION 12:

If the output signal differs from the correct value by a constant, the sensor has an _____.

a. Sensitivity issue
b. offset error
c. drift issue
d. noise

Correct Answer: b. offset error

Detailed Solution: If the output signal differs from the correct value by a constant, the sensor has an offset error or bias.
See lecture 3 @ 21:05

QUESTION 13:

Which of the following is NOT a part of service layer in service-oriented architecture of IoT?

a. Service division
b. Contract
c. Business logic
d. Service repository

Correct Answer: b. Contract

Detailed Solution: Service division, service integration, service repository, business logic, and service bus are the parts of service layer in service-oriented architecture of IoT.
See lecture 5 @ 17:20

QUESTION 14:

Which of the following is the preferred method for dynamic allocation of addresses in an IPv6 network?

a. DHCP
b. SLAC
c. IETF
d. SLAAC/DHCPv6

Correct Answer: d. SLAAC/DHCPv6

Detailed Solution: SLAAC is used to automatically assign an IPv6 address to a host. DHCPv6 is a network protocol for configuring IPv6 hosts with IP addresses, IP prefixes and other configuration data required to operate in an IPv6 network.
See lecture 2 @ 17:20

QUESTION 15:

A proxy-based approach is used to manage multiple IP addresses and map them to _____ addresses.

a. source
b. unique
c. global
d. link local

Correct Answer: d. link local

Detailed Solution: A proxy-based approach is used to manage multiple IP addresses and map them to link local addresses.

See lecture 2 @ 15:35

**********END**********

NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Introduction to Internet of Things

Assignment-Week 2

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15** **Total marks: 15 X 1= 15**

QUESTION 1:

Which of the following is NOT a data and communication protocol?

a. MQTT
b. AMQP
c. Websocket
d. Alljoyn

Correct Answer: d. Alljoyn

Detailed Solution: Alljoyn is an open-source software framework that allows devices to communicate with other devices around them.
See lecture 6 @ 01:00

QUESTION 2:

Which of the following AMQP frame type is used to inform the changes in state of transfer?

a. Flow
b. Disposition
c. Transfer
d. State

Correct Answer: b. Disposition

Detailed Solution: Flow is used to control the message flow rate. Transfer is used for sending actual messages. Disposition is used to inform the changes in state of transfer.
See lecture 8 @ 06:35

QUESTION 3:

Which of the following is NOT a messaging mode of CoAP?

a. Separate
b. Pager
c. Piggyback
d. Confirmable

Correct Answer: b. Pager

Detailed Solution: CoAP has four messaging modes – Confirmable, Non-confirmable, Piggyback, and Separate.
See lecture 7 @ 4:45

QUESTION 4:

The Zigbee Component "APS" stands for _____.

a. Application Support Sub-layer
b. Application Protocol Support
c. Application Policy Support
d. Application Protocol Sub-layer

Correct Answer: a. Application Support Sub-layer

Detailed Solution: APS stands for Application Support Sub-layer. It takes care of interfacing, control services, and bridging between network and other layers.

See lecture 9 @ 18:40

QUESTION 5:

Does IEEE 802.15.4 support IPv6 multicast?

a. Yes
b. No
c. Not Applicable

Correct Answer: b. No

Detailed Solution: IPv6 multicast is not supported by IEEE 802.15.4.
See lecture 10 @ 05:35

QUESTION 6:

In MQTT, a _____ subscribes to a topic.

a. client
b. broker
c. publisher
d. All of the above

Correct Answer: a. client

Detailed Solution: In MQTT, a client/subscriber subscribes to a topic, which is distributed by a broker.
See lecture 6 @ 03:30

QUESTION 7:

Which of the following is a characteristic of XMPP protocol?

a. No interoperability
b. Closed standards
c. No central server
d. All of the above

Correct Answer: c. No central server

Detailed Solution: XMPP is decentralized, based on open standards, and supports interoperability.
See lecture 7 @ 13:30

QUESTION 8:

Which of the following protocol has a component named "bindings"?

a. XMPP
b. AMQP
c. CoAP
d. None of the above

Correct Answer: b. AMQP

Detailed Solution: The three components of AMQP are exchange, queue, and bindings.
See lecture 8 @ 08:10

QUESTION 9:

Is CoAP RESTful?

a. Yes
b. No
c. Not Applicable

Correct Answer: a. Yes

Detailed Solution: The Constrained Application Protocol (CoAP) is a RESTful web transfer protocol for resource-constrained networks and nodes.
See lecture 7 @ 03:50

QUESTION 10:

RPL routing is based upon _____ routing protocol.

a. Enhanced Interior Gateway
b. Interior Gateway

c. Open Shortest Path First
d. Distance vector IPv6

Correct Answer: d. Distance vector IPv6

Detailed Solution: RPL routing is based upon Distance vector IPv6 routing protocol for lossy and low power networks.
See lecture 10 @ 13:45

QUESTION 11:

In general, SMQTT algorithm includes which of the following stages?

a. Publish, decryption & subscribe
b. Encryption, publish, decryption & subscribe
c. Setup, encryption, publish & decryption
d. None of the above

Correct Answer: c. Setup, encryption, publish & decryption

Detailed Solution: SMQTT algorithm consists of 4 main stages - Setup, encryption, publish & decryption.
See lecture 6 @ 13:50

QUESTION 12:

Which of the following modulation technique is used by high data rate versions of IEEE 802.15.4?

a. BPSK
b. QPSK
c. O-QPSK
d. DQPSK

Correct Answer: c. O-QPSK

Detailed Solution: Low speed versions of IEEE 802.15.4 use BPSK. High data rate versions of IEEE 802.15.4 use O-QPSK.
See lecture 9 @ 07:00

QUESTION 13:

The piggyback messaging mode in CoAP Request-Response Model has a “wait” period.

Is the statement true?

a. Yes

b. No
c. Not Applicable

Correct Answer: b. No

Detailed Solution: The separate messaging mode in CoAP Request-Response Model has a "wait" period
See lecture 7 @ 10:30

QUESTION 14:

Which of the following wildcard allows arbitrary values for one hierarchy in MQTT topics?

a. /
b. +
c. #
d. None of the above

Correct Answer: b. +

Detailed Solution: + is a single level wildcard that matches any name for a specific topic level. It only allows arbitrary values for one hierarchy.
See lecture 6 @ 12:25

QUESTION 15:

IEEE 802.15.4 operates in _____ band.

a. LLC
b. SSCS
c. ISS
d. ISM

Correct Answer: d. ISM

Detailed Solution: IEEE 802.15.4 is a standard developed to provide a framework and the lower layers in the OSI model for low cost, low power wireless connectivity networks. It operates in ISM band.
See lecture 9 @ 04:05

**********END**********

NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Introduction to Internet of Things

Assignment-Week 3

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15** **Total marks: 15 X 1= 15**

QUESTION 1:

In general, Z-wave protocol for communication among devices is used for _____.

a. Topology management
b. Home automation
c. Network configuration
d. None of the above

Correct Answer: b. Home automation

Detailed Solution:
See lecture 13 @ 4:05

QUESTION 2:

Which of the following characteristic of HART Data Link Layer helps it achieve collision-free and deterministic communication?

a. Super-frames
b. Channel hopping
c. Channel blacklisting
d. All of the above

Correct Answer: a. Super-frames

Detailed Solution: Collision-free and deterministic communication in HART Data Link Layer is achieved by means of super-frames and TDMA.
See lecture 11 @ 06:40

QUESTION 3:

Which of the following is the basis for Bluetooth technology?

a. Ad-hoc piconets
b. Ad-hoc technology

c. Both (a) and (b)
d. None of the above

Correct Answer: c. Both (a) and (b)

Detailed Solution: Bluetooth technology is based on ad-hoc technology, also known as ad-hoc piconets.
See lecture 12 @ 02:45

QUESTION 4:

Which of the following protocol uses healing (message) to bypass radio dead-spots?

a. HART
b. NFC
c. Z-wave
d. None of the above

Correct Answer: c. Z-wave

Detailed Solution: Z-wave bypasses radio dead-spots using a message called healing.
See lecture 13 @ 10:40

QUESTION 5:

Which of the following nodes send false routing messages in the network?

a. Bad nodes
b. Badly failed nodes
c. Failed nodes
d. All of the above

Correct Answer: b. Badly failed nodes

Detailed Solution: Badly failed nodes send false routing messages which are a threat to the integrity of the network. See lecture 15 @ 04:00

QUESTION 6:

Which of the following layers provide multiplexing capability in Bluetooth technology?

a. L2CAP
b. RFComm
c. SDP
d. Baseband

Correct Answer: a. L2CAP

Detailed Solution: L2CAP provides multiplexing capability in Bluetooth technology.
See lecture 12 @ 12:35

QUESTION 7:

Which of the following component can be considered as optional for a sensor node?

a. Sensing unit
b. Transceiver
c. Location finding unit
d. None of the above

Correct Answer: c. Location finding unit

Detailed Solution: Location finding unit, such as GPS is an optional component for any sensor node.
See lecture 14 @ 08:30

QUESTION 8:

Which of the following provide energy to peer-to-peer tags?

a. Reader
b. Own power source
c. Both (a) and (b)
d. Doesn't require energy

Correct Answer: b. Own power source

Detailed Solution: Active or peer-to-peer tags have their own power source.
See lecture 11 @ 21:10

QUESTION 9:

Which of the following schemes re-establish the connectivity between dumb nodes with others?

a. CoRD
b. CaRAD
c. Both (a) and (b)
d. None of the above

Correct Answer: a. CoRD

Detailed Solution: CoRD and CoRAD are the popular schemes that re-establish the connectivity between dumb nodes with others.

See lecture 15 @ 10:30

QUESTION 10:

Which layer of Bluetooth handles paging?

a. L2CAP
b. RFComm
c. SDP
d. Baseband

Correct Answer: d. Baseband

Detailed Solution: Baseband layer handles packets, paging, and inquiry.
See lecture 12 @ 11:45

QUESTION 11:

In general, is it advisable to increase the duty cycle of a node in case of rare events for social sensing?

a. Yes
b. No
c. Not Applicable

Correct Answer: b. No

Detailed Solution: To monitor rare events for social sensing, usually it is not advisable to increase the duty cycle of a node.
See lecture 15 @ 16:25

QUESTION 12:

Is Zigbee Pro compatible with Zigbee RF4CE?

a. Yes
b. No
c. Not Applicable

Correct Answer: b. No

Detailed Solution: Zigbee, Zigbee Pro, Zigbee RF4CE, and Zigbee IP are incompatible with each other.
See lecture 11 @ 16:45

QUESTION 13:

Which of the following is True?

a. Scatternet is a subset of Piconet

b. Piconet is a subset of Scatternet
c. A device can be a master in more than one piconet
d. A piconet can have maximum of 10 scatternets

Correct Answer: b. Piconet is a subset of Scatternet

Detailed Solution:
See lecture 12

QUESTION 14:

Which of the following suffers in case of total co-operation in wireless ad hoc and sensor networks?

a. Network delay
b. Network throughput
c. Node's lifetime
d. All of the above

Correct Answer: c. Node's lifetime

Detailed Solution: In case of total co-operation in wireless ad hoc and sensor networks, all the relay requests are accepted. This exhausts the limited energy of the nodes.
See lecture 14 @ 28:10

QUESTION 15:

Does ISA 100.11A support tunneling mode?

a. Yes
b. No
c. Not Applicable

Correct Answer: a. Yes

Detailed Solution: A tunneling mode is available to allow legacy data through the ISA 100.11A network.
See lecture 13 @ 16:30

**********END**********

NPTEL Online Certification Courses

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Introduction to Internet of Things

Assignment-Week 4

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 15 **Total marks: 15 X 1= 15**

QUESTION 1:

Which of the following M2M node types have no IP support?

a. Low-end sensor nodes
b. Mid-end sensor nodes
c. High-end sensor nodes
d. All of the above

Correct Answer: a. Low-end sensor nodes

Detailed Solution: Low-end sensor nodes have low capabilities and have no IP support.
See lecture 20 @ 11:00

QUESTION 2:

Which of the following is TRUE for wireless sensor networks if transmission range ≥ 2* sensing range?

a. Only coverage
b. Only connectivity
c. Coverage implies connectivity
d. Connectivity implies coverage

Correct Answer: c. Coverage implies connectivity

Detailed Solution: In WSN, if transmission range ≥ 2* sensing range, coverage implies connectivity.
See lecture 17 @ 04:30

QUESTION 3:

Which of the following structure is usually used in guided formulation of target tracking in sensor networks?

a. Face
b. Tree
c. Cluster

d. All of the above

Correct Answer: a. Face

Detailed Solution: Face structure is often used in guided formulation of target tracking. See lecture 16 @ 06:15

QUESTION 4:

Which of the following is NOT a characteristic of UAV networks?

a. Intermittent link nature
b. Static network topology
c. Dynamic relative position of UAVs
d. None of the above

Correct Answer: b. Static network topology

Detailed Solution: Intermittent link nature, frequent change in network topology, change in relative position of UAV, and lack of suitable routing algorithms are some of the characteristics of UAV networks.
See lecture 19 @ 05:30

QUESTION 5:

Which of the following is a challenge in human-centric sensing?

a. Device lifetime
b. Participant selection
c. Privacy of users
d. All of the above

Correct Answer: d. All of the above

Detailed Solution: Energy of devices, participant selection, and privacy of users are the challenges in human-centric sensing. The energy of the device decides its lifetime.
See lecture 18 @ 12:55

QUESTION 6:

Which of the following components in M2M Service Platform takes care of App management and App searching?

a. Device
b. User
c. Application
d. Access

Correct Answer: d. Access

Detailed Solution: The access control takes care of functionalities, such as app management, app searching, and web portal.
See lecture 20 @ 15:45

QUESTION 7:

In distributed coverage algorithm, nodes compute their position by communicating with _____.

a. Central node
b. Sink node
c. Neighbors only
d. None of the above

Correct Answer: c. Neighbors only

Detailed Solution: In distributed coverage algorithm, nodes compute their position by communicating with their neighbors only.
See lecture 17 @ 10:20

QUESTION 8:

In multi-star UAV network topology, _____ node(s) from each group connects to the ground station.

a. Zero
b. One
c. More than one
d. All

Correct Answer: b. One

Detailed Solution: In multi-star UAV network topology, UAVs form multiple star topology. One node from each group connects to the ground station.
See lecture 19 @ 12:40

QUESTION 9:

Electromagnetic communication in nanonetworks takes place in _____ frequency band.

a. GHz
b. THz
c. Nanometer
d. All of the above

Correct Answer: b. THz

Detailed Solution: Electromagnetic communication takes place in THz frequency band.

See lecture 16 @ 26:40

QUESTION 10:

Which of the following is NOT a property of Mobile Ad Hoc Network?

a. Self-Collaborate
b. Self-Heal
c. Self-Protect
d. Self-Optimize

Correct Answer: a. Self-Collaborate

Detailed Solution: MANET follows self-CHOP properties. CHOP stands for configure, heal, optimize, and protect.
See lecture 18 @ 04:45

QUESTION 11:

Which of the following layers seamlessly integrate IP network with M2M area network in a non-IP based M2M network?

a. Application layer
b. Transport layer
c. Network layer
d. Physical layer

Correct Answer: a. Application layer

Detailed Solution: The application layer seamlessly integrates IP network with M2M area network in a non-IP based M2M network.
See lecture 20 @ 18:30

QUESTION 12:

While maintaining sensing coverage in sensor networks, a crossing is covered if it is in the interior region of _____ one node's coverage disk.

a. At most
b. Only
c. At least
d. More than

Correct Answer: c. At least

Detailed Solution: A crossing is covered if it is in the interior region of at least one node's coverage disk.
See lecture 17 @ 18:05

QUESTION 13:

Which of the following UAV network topology is decentralized?

a. Star
b. Infrastructure mesh
c. Ad hoc mesh
d. None of the above

Correct Answer: c. Ad hoc mesh

Detailed Solution: Ad hoc mesh has no central control center. Infrastructure mesh may have a control center.
See lecture 19 @ 13:45

QUESTION 14:

Can node failure result in partition of networks in mobile wireless sensor networks?

a. Yes
b. No
c. Not Applicable

Correct Answer: b. No

Detailed Solution: Node failure may result in partition of networks in static wireless sensor networks.
See lecture 18

QUESTION 15:

Which of the following is TRUE for high-end sensor nodes in M2M node types?

a. High density deployment
b. Unable to handle multimedia data
c. Mobility is essential
d. All of the above

Correct Answer: c. Mobility is essential

Detailed Solution: High-end sensor nodes in M2M supports low density deployment, handles multimedia data, and supports mobility essentially.
See lecture 20 @ 12:40

**********END**********

NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Introduction to Internet of Things

Assignment-Week 5

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 15 **Total marks: 15 X 1= 15**

QUESTION 1:

Does 6LoWPAN allow interoperability between IEEE802.15.4-based wireless devices, as well as other IP-based devices?

a. Yes
b. No
c. Not Applicable

Correct Answer: a. Yes

Detailed Solution: 6LoWPAN allows interoperability between IEEE802.15.4-based wireless devices, as well as other IP-based devices. 6LoWPAN simply uses a bridge to enable communication between other devices.
Refer to the book: S. Misra, A. Mukherjee, and A. Roy, 2020. Introduction to IoT. Cambridge University Press.

QUESTION 2:

Servo motor works on the principle of _____.

a. Electro-magnetism
b. PWM
c. Magnetism
d. None of these

Correct Answer: b. PWM

Detailed Solution: Servo motor works on PWM (Pulse width modulation) principle, means its angle of rotation is controlled by the duration of applied pulse to its Control PIN.
See lecture 25

QUESTION 3:

Which of the following is NOT a commonly available Arduino boards in the market?

a. Arduino Uno
b. Arduino Mega

c. Arduino Nano
d. Arduino Dyu

Correct Answer: d. Arduino Dyu

Detailed Solution: Some of the commonly available Arduino boards in the market are Arduino UNO, Arduino Mega, Arduino Due, and Arduino Nano.

QUESTION 4:

Different processing logics applied to same IoT networked devices or applications is classified as _____.

a. Semantic conflict
b. Configuration conflict
c. Processing conflict
d. Heterogeneity conflict

Correct Answer: a. Semantic conflict

Detailed Solution: Semantic conflict refers to different processing logics applied to same IoT networked devices or applications.
See lecture 21 @ 05:10

QUESTION 5:

Which of the following is TRUE for collaborative conceptualization theory?

a. overcomes the limitation of ontology
b. enables cross-context semantic interoperability for any objects
c. Both overcomes the limitation of ontology and enables cross-context semantic interoperability for any objects
d. represents things as symbols of semiotics.

Correct Answer: c. Both overcomes the limitation of ontology and enables cross-context semantic interoperability for any objects

Detailed Solution: To overcome the limitation of ontology and enable cross-context semantic interoperability for any objects, the collaborative conceptualization theory was developed. These concepts represent things as signs of semiotics.
See lecture 21 @ 20:45

QUESTION 6:

Which of the following is TRUE for the following command?

```
Serial.begin(9600);
```

a. Arduino exchange messages with the Serial Monitor at a data rate of 9600 bytes per second.
b. 9600 represent binary ones or zeros per second.

c. Both Arduino exchange messages with the Serial Monitor at a data rate of 9600 bytes per second and 9600 represent binary ones or zeros per second.
d. Arduino exchange messages with the Serial Monitor for 9600 simulation time.

Correct Answer: b. 9600 represent binary ones or zeros per second

Detailed Solution: As per the basics of Arduino programming.
See lecture 22

QUESTION 7:

Which of the following standards can be utilized as Device Catalogs?

a. ecl@ss
b. UNSPSC
c. Both UNSPSC and ecl@ss
d. None of these

Correct Answer: c. Both UNSPSC and ecl@ss

Detailed Solution: There are two existing standards on commodity description and classification, which are UNSPSC (www.unspsc.org) and ecl@ss (www.eclass.de). These two standards can be introduced to IoT for device categorization purpose.
See lecture 21 @ 16:25

QUESTION 8:

Which of the following enables interoperability?

a. Cloud computing system
b. Sensor nodes
c. Middleware
d. All of these

Correct Answer: c. Middleware

Detailed Solution: Interoperability is enabled by middleware.
See lecture 21

QUESTION 9:

Which of the following is TRUE for the sketch command given below?

```
delay(1000);
```

a. Provides a delay of 1000 seconds
b. Provides a delay of 1 second
c. Provides a delay of 1000 simulation time
d. Provides a delay of 1 simulation time

Correct Answer: b. Provides a delay of 1 second

Detailed Solution: As per the basics of Arduino programming.

See lecture 22

QUESTION 10:

In a sketch, a string can be declared using _____.

a. Array
b. String object
c. Both array and string object
d. None of these

Correct Answer: c. Both array and string object

Detailed Solution: As per the basics of Arduino programming.
See lecture 23

QUESTION 11:

In syntactic interoperability between devices and device user, the message format from the user to the device is _____ by the device.

a. understandable
b. executable
c. Both understandable and executable
d. None of the above

Correct Answer: b. Executable

Detailed Solution: In syntactic interoperability between devices and device user, the message format from the user to the device is executable by the device. While, the message format from a device to a user is understandable for the user's computer.
See lecture 21 @ 16:55

QUESTION 12:

In a sketch, which of the following function initializes I/O variables and pin modes?

a. initialize()
b. loop()
c. init()
d. setup()

Correct Answer: d. setup()

Detailed Solution: As per the basics of Arduino programming.
See lecture 22

QUESTION 13:

Which does pin 3 (from left to right, as shown in the figure below) in a digital humidity and temperature sensor signify?

a. Power supply
b. Data
c. Ground
d. None of these

Correct Answer: d. None of these

Detailed Solution: As per the sensor basics.
See lecture 24

QUESTION 14:

Which of the following best describes the command given below?

ServoDemo.write(180);

a. Creates an instance of the servo
b. Pin writes 180 to the servo
c. Servo moves 180 degrees
d. All of these

Correct Answer: c. Servo moves 180 degrees

Detailed Solution: As per the basics of Arduino libraries.
See lecture 25 @ 08:15

QUESTION 15:

Which of the following is an interoperable system commonly used for automation?

a. LonWorks
b. Dropbox
c. ThingSpeak
d. All of these

Correct Answer: a. LonWorks

Detailed Solution: LonWorks® is an open and interoperable system for building automation and is characterized by its flexible topology and cross-system functions.
Refer to the book: S. Misra, A. Mukherjee, and A. Roy, 2020. Introduction to IoT. Cambridge University Press.

**********END**********

NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Introduction to Internet of Things

Assignment-Week 6

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 15 **Total marks: 15 X 1= 15**

QUESTION 1:

What is the output for the following piece of Python code?

```
x = [32, 'u', 'i', 8, '34']
x = x[0:]
print (x)
```

a. [32]
b. ['34']
c. ['u', 'i', 8, '34']
d. [32,'u', 'i', 8, '34']

Correct Answer: d. [32, 'u', 'i', 8, '34']
Detailed Solution: As per program output.

QUESTION 2:

How many GPIO (General Purpose Input Output Pin) pins are there in Raspberry Pi 4?

a. 30
b. 14
c. 40
d. 41

Correct Answer: c. 40

Detailed Solution: All the Raspberry Pi models have 40 GPIO pins.

QUESTION 3:

Which of the following is NOT an example of Python IDE?

a. Sublime Text
b. PyCharm
c. Spider
d. None of the above

Correct Answer: c. Spider

Detailed Solution: Sublime Text, PyCharm, Spyder, and Jupyter are few of the examples of Python IDE.
See lecture 26

QUESTION 4:

What is the value of 'x' in the following expression in Python programming?

x = 3^3

a. 0
b. 9
c. -2
d. Will raise an exception

Correct Answer: a. 0

Detailed Solution: As per program output. "^" is a bitwise XOR operator in python.

QUESTION 5:

In python programming, which of the following is a null statement?

a. Pass
b. Continue
c. Break
d. Skip

Correct Answer: a. Pass

Detailed Solution: As per basics of Python programming, pass is a null operation. The interpreter does not ignore a pass statement, but nothing happens and the statement results into no operation.

QUESTION 6:

Which of the following is an unordered data type in Python?

a. List
b. Dictionary
c. Both List and Dictionary
d. Tuple

Correct Answer: b. Dictionary

Detailed Solution: As per basics of Python programming.
See lecture 26 @ 15:36

QUESTION 7:

What is the value of 'x' in the following expression in Python programming?

x = 2**3^3**1

a. 11
b. 0
c. Will raise an error
d. Will raise an exception

Correct Answer: a. 11

Detailed Solution: As per program output

QUESTION 8:

Which of the following symbol is used to comment out multiple lines at once in python?

a. $
b. #
c. %
d. None of these

Correct Answer: d. None of these

Detailed Solution: Unlike other programming languages Python doesn't support multi-line comment blocks out of the box. The recommended way to comment out multiple lines of code in Python is to use consecutive # single-line comments.

QUESTION 9:

With on-board Raspberry Pi camera, which one of the following is NOT correct for image related operations in Python?

a. from PIcamera import PiCamera
b. from picamera import PyCamera
c. from pycamera import PiCamera
d. All of these

Correct Answer: d. All of these

Detailed Solution: As per program output, the correct statement is "from picamera import PiCamera"

QUESTION 10:

Which of the following bit processor is used in Raspberry Pi Zero 2 W?

a. 64
b. 32

c. Both 64 & 32
d. 128

Correct Answer: a. 64

Detailed Solution: The Raspberry Pi Zero 2 W includes a new Broadcom BCM2710A1 SOC providing a 1GHz quad-core 64-bit ARM Cortex-A53 CPU and 512MB RAM.

QUESTION 11:

Which of the following shortcut exits the nano editor?

a. Ctrl + E
b. Ctrl + O
c. Ctrl + X
d. None of these

Correct Answer: c. Ctrl + X

Detailed Solution: Ctrl + O writes the code to a file. Ctrl + X exits the nano editor.
See lecture 29

QUESTION 12:

What of the following syntax is correct for networking in python?

a. s = socket.socket(socket.AF_NET, socket.SOCK_STRAM)
b. s = socket.socket(socket.AF_UNIX, socket.SOCK_DGRAM)
c. s = socket.socket(socket.AF_UNX, socket.SOCK_TCP)
d. s = socket.socket(socket.AF_INET, socket.SOCK_UDP)

Correct Answer: b. s = socket.socket(socket.AF_UNIX, socket.SOCK_DGRAM)

Detailed Solution: AF_UNIX or AF_INET represent socket family. SOCK_STREAM or SOCK_DGRAM represent socket type.
See lecture 27

QUESTION 13:

Which of the following is correct to open a text file for write mode?

a. open('file.txt', ‘w’)
b. open('file.txt', ‘+w’)
c. open('file.txt', ‘w+’)
d. open('file.txt', ‘o+w’)

Correct Answer: a. open('file.txt', ‘w’)

Detailed Solution: ‘w’ mode is for writing into a text file.
See lecture 27 @ 02:50

QUESTION 14:

Which of the following is an immutable data type in Python?

a. String
b. Tuple
c. Bool
d. All of these

Correct Answer: d. All of these

Detailed Solution: Immutable Objects are of in-built types like int, float, bool, string, unicode, tuple. In simple words, an immutable object can't be changed after it is created.

QUESTION 15:

Does python follow rigid indentation?

a. Yes
b. No
c. Not Applicable

Correct Answer: a. Yes

Detailed Solution: As per basics of Python programming.

**********END**********

NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Introduction to Internet of Things

Assignment-Week 7

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 15 **Total marks: 15 X 1= 15**

QUESTION 1:

In Socket programming, the parameter AF_INET stands for _____.

a. Unix protocols
b. Internet Protocol (IP)
c. File sharing
d. Time slicing

Correct Answer: b. Internet Protocol (IP)

Detailed Solution The AF_INET specifies the rules and standards of the Internet protocol, hence the socket acts as an IP socket. (Please refer Lecture 31@14:13)

QUESTION 2:

During remote server access by a Raspberry Pi, where the Raspberry Pi acts as a client, the client needs the following?

a. Only IP address of server
b. Only port number
c. Both server IP address and port number
d. Client's IP address

Correct Answer: c. Both server IP address and port number

Detailed Solution: A client can communicate with a server only if both IP address and port numbers are known. (Please refer Lecture 31@14:13)

QUESTION 3:

If you are using Raspberry Pi to connect IoT sensors (such as DHT temperature sensors), which of the following python-based libraries lets you configure the sensors and read data from them?

a. Adafruit
b. Matplotlib
c. Numpy
d. Socket

Correct Answer: a. Adafruit

Detailed Solution: The Adafruit library provides the necessary functions to configure sensors and read data from them to a Raspberry Pi. (Please refer Lecture 31@4:45)

QUESTION 4:

You can store and log the different sensor data for processing in text (.txt) files but not comma separated value (.csv) files.

a. True
b. False

Correct Answer: b. False

Detailed Solution: The sensor data can be logged in any format as is suitable and appropriate, as long the data does not get corrupted. (Please refer Lecture 32 full and @26:14 and 25:43)

QUESTION 5:

Consider the following python script using the split() function, what will be the correct print output (SEE the options VERY carefully including the quotation marks)

```
dat = 'Apple,Guava#Banana'
var = dat.split(""#"")
print(var)
```

a. ['Apple','Guava','Banana']
b. ['Apple','Guava']
c. ['Guava','Banana']
d. ['Apple,Guava','Banana']

Correct Answer: d. ['Apple,Guava','Banana']

Detailed Solution: The split('#') method splits the string into two parts with respect to the '#' character**.** (Please refer Lecture 32@8:02)

QUESTION 6:

In SDN, the central controller defines all the paths, flows and behavior of the network in a centralized manner.

a. True
b. False

Correct Answer: a. True

Detailed Solution: SDN is a centralized networking scheme where the central controller has full global control over the network. (Please refer Lecture 33@14:47)

QUESTION 7:

OpenFlow is a protocol that is used for traditional distributed OSPF based networking.

a. True
b. False

Correct Answer: b. False

Detailed Solution: OpenFlow is the most popular protocol that implements concepts of SDN, where SDN is the theoretical framework and is different from traditional networks (Please refer Lecture 33@24:20)

QUESTION 8:

In SDN _____ tables in switches describe the rules that the switch must follow for packet forwarding and are configured by the controller.

a. network
b. distance
c. hop
d. flow
.

Correct Answer: d. flow

Detailed Solution: Flow tables are configured in SDN switches by the controller that determine its packet forwarding behavior. (Please refer Lecture 33@17:26)

QUESTION 9:

In a network architecture one logical hop between two nodes can actually be composed of multiple physical hops in the physical network.

a. True
b. False

Correct Answer: a. True

Detailed Solution: A logical connection between two nodes can actually contain several physical hops (Please refer Lecture 34@4:35)

QUESTION 10:

What among the following is a challenge while designing SDN architecture?

a. Control Placement only
b. Both Rule and control placement
c. Rule placement only
d. Switch placement.

Correct Answer: b. Both Rule and control placement

Detailed Solution: Both rule and control placements are issues in SDN (Please refer Lecture 33@16:52)

QUESTION 11:

PACKET_IN type of messages are associated with which of the following directional APIs?

a. Eastbound APIs
b. Southbound APIs
c. Westbound APIs
d. Northbound APIs

Correct Answer: b. Southbound APIs

Detailed Solution: Southbound APIs are for communication between the controller and the switches and PACKET_IN is used by the switches to send a new packet to the controller. (Please refer Lecture 33@22:31)

QUESTION 12:

For an SDN switch flow rule, the soft timeout is set to 2 minutes and the hard timeout is set to 6 minutes. After 6 minutes and 10 seconds since the flow rule was initialized, what will be the status of the flow be in the switch?

a. The flow will be surely deleted
b. The flow will be deleted if it receives no packet for 4 minutes.
c. The flow will remain intact.
d. The flow will remain intact with a random probability.

Correct Answer: a. The flow will be surely deleted

Detailed Solution: Hard timeout signifies the time after which the flow rule will no longer be valid in any situation. (Please refer Lecture 33@24:51)

QUESTION 13:

It is a good idea to keep a backup controller in an SDN architecture along with the main controller.

a. True
b. False

Correct Answer: a. True

Detailed Solution: The backup controller is useful for storage and recovery in case the primary controller fails. It makes the network resilient. (Please refer Lecture 34@8:30)

QUESTION 14:

Software Defined IoT solves the challenge of _____ that is prevalent in traditional IoT networks.

a: uniformity
b: deployment
c: homogeneity
d: heterogeneity

Correct Answer: d. heterogeneity

Detailed Solution: SDIoT can be used to solve the prevalent heterogeneity challenges in traditional IoT networks. (Please refer Lecture 35@6:33)

QUESTION 15:

Sensor OpenFlow, Soft-WSN and SDN-WISE are examples of _____.

a. Traditional routing protocols.
b. Various SDN protocols for wired LANs
c. Various implementations of the concept of Software Defined IoT
d. Traditional IoT protocol examples.

Correct Answer: c. Various implementations of the concept of Software Defined IoT

Detailed Solution: The given names are all examples of various implementations of different concepts of Software Defined IoT by various research groups across the world. (Please refer Lecture 35@8:46 onwards)

**********END**********

NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Introduction to Internet of Things

Assignment-Week 8

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15** **Total marks: 15 X 1= 15**

QUESTION 1:

Which of the following is an issue prevalent for communication in wireless sensor networks?

a. Diffusion
b. Interference
c. Diffraction
d. Inheritance

Correct Answer: b. Interference

Detailed Solution In any wireless networks that communicate using radio frequency, interference is a major issue. (Please refer Lecture 36@4:27)

QUESTION 2:

In a cloud computing paradigm, the user does not have availability to on-demand services.

a. True
b. False

Correct Answer: b. False

Detailed Solution: Cloud computing is built for providing dynamic on-demand services (Please refer Lecture 37@9:51 onwards)

QUESTION 3:

Which among the following can be rented as a utility service by a client remotely in Cloud Computing?

a. Hardware
b. Software
c. Memory and storage
d. All of these

Correct Answer: d. All of these

Detailed Solution: Cloud computing provides clients on-demand services to all computing features, from hardware to software to infrastructure. (Please refer Lecture 37@9:15)

QUESTION 4:

When you are accessing Google docs or Microsoft Word online for accessing the features of a document writing software from your browser without specifically installing them, which cloud service model you are using?

a. SaaS
b. PaaS
c. IaaS
d. DaaS

Correct Answer: a. SaaS

Detailed Solution: This is an example of SaaS, since you are accessing a word/document processing software as a client through the client. (Please refer Lecture 37@26:14 AND 38@13:16)

QUESTION 5:

In cloud computing service model, what does 'XaaS' stand for?

a. Security-as-a-Service
b. Connectivity-as-a-Service
c. Network-as-a-Service
d. Anything-as-a-Service

Correct Answer: d. Anything-as-a-Service

Detailed Solution: 'X' in XaaS stands for anything. (Please refer Lecture 37@25:00)

QUESTION 6:

Microsoft Azure is an example of _____ service provider.

a. SDN
b. Cloud computing
c. Batch computing
d. Operating System

Correct Answer: b. Cloud computing

Detailed Solution: Azure is the cloud service by Microsoft (Please refer Lecture 40@21:38)

QUESTION 7:

No virtualization is used in cloud service provisioning**.**

a. True
b. False

Correct Answer: b. False

Detailed Solution: Cloud computing uses the concept of virtualization heavily to provide continuous, dynamic on-demand services. (Please refer Lecture 37 through 40, specially for PaaS and IaaS. This concept has been covered multiple times)

QUESTION 8:

What does DBaaS stand for in cloud computing service model?

a. Disk-as-a-Service.
b. Database-as-a-Service
c. Distribution-as-a-Service
d. Data buffering-as-a-Service
.

Correct Answer: b. Database-as-a-Service

Detailed Solution: DBaaS stands for Database-as-a-Service

(Please refer Lecture 39@13:32)

QUESTION 9:

Suppose a hospital that deals with sensitive patient data wants to have cloud support for all of its computers, but it wants to set up its own secure cloud infrastructure. This is an example of which of the following models?

a. Public cloud
b. Hybrid cloud
c. Private cloud
d. None of these

Correct Answer: c. Private cloud

Detailed Solution: As the computers can access the cloud set up by the hospital's own infrastructure, this is an example of private cloud (Please refer Lecture 39@14:46)

QUESTION 10:

What does 'CIA' in cloud data security stand for?

a. Confidentiality, Integrity, Availability
b. Confidentiality, Inheritance, Automation
c. Congestion, Integrity, Authentication
d. Criticality, Integrity, Accountability

Correct Answer: a. Confidentiality, Integrity, Availability

Detailed Solution: 'CIA' stands for 'Confidentiality, Integrity and Availability'. (Please refer Lecture 39@21:01)

QUESTION 11:

In virtualization, a hypervisor is responsible for orchestration between the underlying physical hardware and the virtual machines being installed.

a. True
b. False

Correct Answer: a. True

Detailed Solution: Hypervisor provides the necessary orchestration between virtual machines and host computers. (Please refer Lecture 38@5:50)

QUESTION 12:

For a secure cloud deployment, which among the following are needed to be considered?

a. Network Level Security
b. Host Level Security
c. Application Level Security
d. All of these

Correct Answer: d. All of these

Detailed Solution: Network, Host and Application level security are all important. (Please refer Lecture 39@17:00)

QUESTION 13:

CloudSim is a type of

a. Cloud network
b. Cloud simulator
c. Cloud enhancement
d. Cloud model

Correct Answer: b. Cloud simulator

Detailed Solution: CloudSim is a type of cloud simulator. (Please refer Lecture 40@4:47)

QUESTION 14:

Before cloud computing provisioning, the client is provided with the details of the various warranties in which of the following document?

a. Service Level Authentication
b. Service Level Agreement
c. Software Level Agreement
d. Software Level; Authentication

Correct Answer: b. Service Level Agreement

Detailed Solution: SLA stands for Service Level Agreement. It defines all the service level requirements and promises bly the cloud service provider. (Please refer Lecture 39@4:36)

QUESTION 15:

Amazon Web Services (AWS) is an example of

a. Open source cloud
b. Deployment cloud
c. Commercial cloud
d. Not a cloud platform

Correct Answer: c. Commercial cloud

Detailed Solution: AWS is the commercial cloud service by Amazon. (Please refer Lecture 40@18:37)

**********END**********

NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Introduction to Internet of Things

Assignment-Week 9

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 15 **Total marks: 15 X 1= 15**

QUESTION 1:

OpenStack is proprietary and a closed source software that you have to pay and buy.

a. True
b. False

Correct Answer: b. False

Detailed Solution: OpenStack is open source and requires no payment. (Please refer Lecture 41@2:09)

QUESTION 2:

In OpenStack which of the following component takes care of user authentication?

a. Nova
b. Horizon
c. Keystone
d. None of these

Correct Answer: c. Keystone

Detailed Solution: Keystone component in OpenStack provides authentication and user authorization features (Please refer Lecture 41@4:07)

QUESTION 3:

You cannot ping your OpenStack instance from an outside network unless you connect your instance with the public network through a _____

a. Router
b. Firewall
c. Repeater
d. Load balancer

Correct Answer: a. Router

Detailed Solution: You are supposed to connect your virtual OpenStack instance with the public network through a router (Please refer Lecture 41@17:41)

QUESTION 4:

What is the primary service model for a sensor cloud?

a. Hardware-as-a-Service
b. Platform-as-a-Service
c. Data-as-a-Service
d. Sensor-as-a-Service

Correct Answer: d. Sensor-as-a-Service

Detailed Solution: The primary functionality of a sensor cloud is to provide Sensor-as-a-Service. (Please refer Lecture 42@1:00)

QUESTION 5:

You can run two logical computers, one Linux machine and one Windows machine on top of the same physical hardware through the use of _____.

a. Sharing
b. Scaling
c. Virtualization
d. Inheritance.

Correct Answer: c. Virtualization

Detailed Solution: Virtualization supports multiple logical instances of different computers on the same hardware. (Please refer Lecture 42@8:50)

QUESTION 6:

In Sensor Cloud, one physical sensor can be abstracted as multiple virtual sensors.

a. True
b. False

Correct Answer: a. True

Detailed Solution: Virtualization allows the abstraction of one physical sensors into multiple virtual sensors. (Please refer Lecture 42@12:10)

QUESTION 7:

In a Sensor cloud framework, the end-users have to continuously worry about owning and managing the physical sensor devices.

a. True
b. False

Correct Answer: b. False

Detailed Solution: Sensor cloud supports the existence of multiple actors with end-users concerned only with the application data and not internal details. (Please refer Lecture 42@13:09)

QUESTION 8:

Virtual sensors within a sensor cloud have communication interfaces with _____.

a. Only the physical sensors below them
b. Only the end-user applications above them.
c. Both physical sensors below and applications above.
d. Neither the physical sensors, nor the applications above.

Correct Answer: c. Both physical sensors below and applications above

Detailed Solution: Virtual sensor layer sits in the middle having interface to both physical sensors as well as applications. (Please refer Lecture 43@9:24 and 28:01)

QUESTION 9:

Which of the following is true?

a. Traditional WSN: Virtualization support, Sensor Cloud: Virtualization support
b. Traditional WSN: No virtualization, Sensor Cloud: Virtualization Support
c. Traditional WSN: Virtualization support, Sensor Cloud: No Virtualization
d. Traditional WSN: No virtualization, Sensor Cloud: No Virtualization

Correct Answer: b. Traditional WSN: No virtualization, Sensor Cloud: Virtualization Support

Detailed Solution: Sensor Cloud supports complete virtualization (Please refer Lecture 42@12:04)

QUESTION 10:

An organization ABC wishes to have environmental monitoring parameters of a particular region, but does not have the resources to deploy its own physical sensors. However, another organization B has some physical sensors that monitor the environment deployed in that region. Can sensor cloud help in achieving A's objectives?

a. Yes
b. No

Correct Answer: a. Yes

Detailed Solution: Sensor cloud can provide virtual sensor services to A (Please refer Lecture 42@17:57)

QUESTION 11:

Referring back to Question 10 (above) and based on your answer, determine the roles that organization A and B play in the given scenario.

a. A: Sensor owner, B: End-user (service consumer)
b. A: End-User (service consumer), B: Sensor owner
c. Both A and B are sensor owners
d. Both A and B are end-users

Correct Answer: b. A: End-User (service consumer), B: Sensor owner

Detailed Solution: As B owns the physical sensors and has deployed them, B is the sensor-owner and as A requires the sensor services, it is the end-user (Please refer Lecture 42@17:57)

QUESTION 12:

Which among the following is a limitation of cloud computing?

a. On-demand services
b. Virtualization orchestration
c. Large amount of computing and storage resources.
d. Additional communication latency for critical applications

Correct Answer: d. Additional communication latency for critical applications

Detailed Solution: Communication latency to and from cloud is one of the major drawback for clouds, especially for critical applications (Please refer Lecture 44@1:01)

QUESTION 13:

In Fog computing, the Fog layer acts as an intermediary between the data source and the cloud for immediate processing of time critical data.

a. True
b. False

Correct Answer: a. True

Detailed Solution: Fog actually sits in between the device and cloud to provide immediate services (Please refer Lecture 44@7:29 and Lecture 45)

QUESTION 14:

Suppose the one-way communication latency from a device to a Fog layer is 20 ms and that from the device to the cloud layer is 50ms. Data processing time at Fog is 10ms and that at Cloud is 5 ms, after which (i.e completion of data processing) a reply is sent back to the device by the Fog or Cloud. What is the difference between total communication latency (two-way round-trip communication plus processing time at Fog/Cloud) between the device and the Cloud and Fog respectively?

a. 50ms
b. 60ms
c. 55ms
d. 100ms

Correct Answer: c. 55ms

Detailed Solution: Total communication latency for cloud = 50 + 5 + 50 = 105ms. Total communication latency for Fog = 20 + 10 + 10 = 50ms. Therefore difference = 105-50 = 55 ms (Please refer Lecture 45@5:06)

QUESTION 15:

Which among the following is true?

a. Fog computing acts as a complement to cloud computing.
b. Fog computing is a replacement for cloud computing.
c. Fog computing and cloud computing are the same.
d. Fog computing is more powerful than cloud computing (with respect to resources).

Correct Answer: a. Fog computing acts as a complement to cloud computing.

Detailed Solution: Fog computing and cloud computing are complementary technologies. Fog helps in bringing the cloud closer to the IoT devices. (Please refer Lecture 45@1:06)

**********END**********

NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Introduction to Internet of Things

Assignment-Week 10

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 15 **Total marks: 15 X 1= 15**

QUESTION 1:

Fill in the blank. Facilities that need to be provided in smart parking lot includes _____.

a. Identification of empty slots in parking slot
b. Automatic routing of vehicles in vacant slot
c. Automatic charging for services rendered
d. All of these

Correct Answer: d. All of these

Detailed Solution: Current focus of area related to smart parking lot includes identification of empty slots in parking slot, automatic routing of vehicles in vacant slot, automatic charging for services rendered. (Please refer Lecture 46@25:10)

QUESTION 2:

In the near future, auto alert generation in case of sudden heart attack leading to emergency medical condition is supposed to be handled by smart health.

a. True
b. False

Correct Answer: a.

Detailed Solution: Current focus area in smart home includes auto alert generation in case of emergency medical condition (e.g., sudden heart attack) (Please refer Lecture 46@27:00)

QUESTION 3:

During an event of disaster, the resource reallocation and service rerouting can be handled by which of the following method?

a. Energy monitoring
b. Traffic monitoring
c. Calamity monitoring
d. Health monitoring

Correct Answer: c. Calamity monitoring

Detailed Solution: The resource reallocation and service rerouting in the event of calamity is performed through pollution and calamity monitoring. (Please refer Lecture 46@27:31)

QUESTION 4:

Which of the following challenges can be experienced due to the periodic generation of the large volume of data in smart city?

a. Making data volume precise and accurate
b. Provisioning quality of service
c. Proper resource management
d. None of these

Correct Answer: a. Making data volume precise and accurate

Detailed Solution: Enormous volume of data is produced periodically in a smart city environment. Challenges includes making the available or incoming data volume precise and accurate. (Please refer Lecture 47 @5:21)

QUESTION 5:

Fill in the blank.

_____ enables optimal usages of the huge data generated from different sources and across different platforms.

a. Data storage
b. Data cleaning
c. Data fusion
d. None of these

Correct Answer: c. Data fusion

Detailed Solution: Data fusion enables optimum utilization of the massive data generated from different sources and across different platforms. (Please refer Lecture 47 @6:15)

QUESTION 6:

Which of the following is included in mathematical model that is used for performing data fusion?

a. Convolutional neural network neural network
b. Statistical model
c. Belief function
d. All of these

Correct Answer: d. All of these

Detailed Solution: Mathematical model of data function includes convolutional neural network neural network, statistical model, belief function, machine learning, artificial neural network, etc. (Please refer Lecture 47@11:49)

QUESTION 7:

Which of the following statements are true about the different components present in a smart home?

Statement I: Sensor networks and communication network infrastructure

Statement II: Intelligent control and management

Statement III: Manual instead of automatic access of devices

a. Statements I and II
b. Statements I and III
c. Statements II and III
d. Statements I, II and III

Correct Answer: a. Statements I and II

Detailed Solution: The different components of smart home include communication network infrastructure, Intelligent control and management, Sensor networks, smart features and automatic response. (Please refer Lecture 48@7:04)

QUESTION 8:

Which of the following are included in the HAN standards?

Statement I: UPnP and LonWorks

Statement II: Zigbee and X-10

Statement III: DLNA and Zigbee

a. Statement I and II
b. Statement II and III
c. Statement I and III
d. Statement I, II and III

Correct Answer: d. Statement I, II and III

Detailed Solution: HAN standards include UPnP, LonWorks, DLNA, Zigbee and X-10. (Please refer Lecture 48@12:53)

QUESTION 9:

Fill in the blank.

_____ provisions amalgamation of different features at home including home automation system, personal media, security system, etc.

a. Local area network
b. Personal area network
c. Metropolitan area network
d. Home area network

Correct Answer: d. Home area network

Detailed Solution: Home area network provides amalgamation of various features at home such as- home automation system, personal media, security system, etc. (Please refer Lecture 48@8:15)

QUESTION 10:

Fill in the blank. In Vehicle-to-human / human-to-vehicle interaction, the human communicating with the vehicle _____.

a. Is present in another vehicle

b. Is present in the same vehicle

c. Is present outside the vehicle on the roadside

d. All of these

Correct Answer: c. Is present outside the vehicle on the roadside

Detailed Solution: In V2H/H2V, the human is present outside the vehicle on the roadside. (Please refer Lecture 49@3:13)

QUESTION 11:

Information centric networks are derived from content centric network architecture.

a. True
b. False

Correct Answer: b. False

Detailed Solution: Content centric network architecture is derived from information centric networks. (Please refer Lecture 49@15:55)

QUESTION 12:

Which of the following statement(s) is/are true about the Vehicular Ad-hoc network (VANET)?

a. Routing protocol of MANET is derived from VANET
b. High latency is achieved in mobile environment
c. Does not have wireless access in vehicular environment
d. None of these

Correct Answer: d. None of these

Detailed Solution: In Vehicular Ad-hoc network, routing protocol of VANET is derived from MANET, high throughput is achieved in mobile environment, and it is based on wireless access in vehicular environment. (Please refer Lecture 49@16:41)

QUESTION 13:

Which of the following is considered in Intelligent connected vehicles?

a. Intelligent transportation
b. Communication channels
c. Pedestrian
d. All of these

Correct Answer: d. All of these

Detailed Solution: Intelligent connected vehicles consider intelligent transportation, pedestrian, communication channels, and transportation infrastructure (Please refer Lecture 50@7:06)

QUESTION 14:

Dedicated short range communication standards developed by Society of Automotive Engineers are SAE J2735 and J2945.

a. True
b. False

Correct Answer: a. True

Detailed Solution: Society of Automotive Engineers came up with SAE J2735 and J2945 as DSRC standards (Please refer Lecture 50@8:49)

QUESTION 15:

What are the disadvantages of V2X communication?

a. Increased traffic safety
b. Efficient use of fuel
c. Tracking of movement
d. None of these

Correct Answer: c. Tracking of movement

Detailed Solution: Disadvantages of V2X communication includes tracking of movement, violation of privacy, loss of data control, etc. (Please refer Lecture 50@21:04)

**********END**********

NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Introduction to Internet of Things

Assignment-Week 11

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15** **Total marks: 15 X 1= 15**

QUESTION 1:

Which of the following is true about traditional electrical grid?

a. Monitoring and restoration is done automatically
b. Energy is distributed from plant to the home or industries
c. Energy generation is done in distributed power plants
d. Bi-directional communication

Correct Answer: b. Energy is distributed from plant to the home or industries

Detailed Solution: In traditional grid, energy distribution is one directional - from plant to the home or industries (Please refer Lecture 51@3:08)

QUESTION 2:

Power monitoring along with asset management and optimal utilizations falls under which characteristics of smart grid?

a. Power system efficiency
b. Real time pricing
c. Distributed generation
d. Power quality

Correct Answer: a. Power system efficiency

Detailed Solution: Power monitoring, asset management and optimal utilizations, distribution automation and protection falls under the power system efficiency properties of smart grid (Please refer Lecture 51@22:07)

QUESTION 3:

Which of the following statements are true about net metering?

Statements:

I. Installation of smart meter does not contribute to net metering

II. Consumers are speed high on supply in excess amount of generated energy to do the grid during on-peak hours.

III. In case of off-peak hours, the price is less.

a. Statements I and II

b. Statements I and III

c. Statements II and III

d. Statements I, II, and III

Correct Answer: c. Statements II and III

Detailed Solution: Firstly, consumers are paid high, if they are supplying excess amounts of generated energy to the grid in on-peak hours. Secondly, the price is less in case of off-peak hours. (Please refer Lecture 51@35:07)

QUESTION 4:

Which of the following is true about plug-in electric vehicles?

a. PEV does not rely on power plants to charge their batteries.

b. The consumer is discouraged by energy service providers to charge their batteries of PEV during off peak hours.

c. By using electricity, the PV does not reduce pollution.

d. PEV can be utilized as an energy source during on-peak hours.

Correct Answer: d. PEV can be utilized as an energy source on peak hours

Detailed Solution: PEV also can be utilized as an energy source on peak hours. (Please refer Lecture 52@9:42)

QUESTION 5:

What are the advantages of traditional operation centers?

a. Tries to confirm that the amount of energy generated is utilized
b. Grid voltage drop due to excessive current generation leads to unsteady grid
c. Controlling capabilities are not boundless
d. None of these

Correct Answer: d. None of these

Detailed Solution: The above-mentioned points (a, b, and c) are the drawbacks of traditional operation centers. (Please refer Lecture 52@1:44)

QUESTION 6:

Fill in the blank. Avoiding unwanted current flow through grid is tackled by _____.

a. Demand side energy distribution by the grid
b. Self-healing capability of the grid
c. Energy serving in specific geographical location in smart grid
d. All of these

Correct Answer: b. Self-healing capability of the grid.

Detailed Solution: Self-healing capability of the grid avoid unwanted flow of current through the grid (Please refer Lecture 52@3:40)

QUESTION 7:

Select the correct order of evolution of industry 4.0.

a. Power generation and mechanical automation, smart automation, electronic automation, industrialization

b. Power generation and mechanical automation, electronic automation, industrialization, smart automation

c. Industrialization, power generation and mechanical automation, electronic automation,

smart automation

d. Power generation and mechanical automation, industrialization, electronic automation, smart automation

Correct Answer: d. Power generation and mechanical automation, industrialization, electronic automation, smart automation

Detailed Solution: Power generation and mechanical automation (1782), industrialization (1870), electronic automation (1969), smart automation (today). (Please refer Lecture 53@10:07)

QUESTION 8:

Industrial Internet of Things (IIoT) is a network of physical objects, systems, platforms and application. These networks can interact with each other, external environment but not with other people.

a. True
b. False

Correct Answer: b. False

Detailed Solution: Industrial Internet of Things (IIoT) is a network of physical objects, systems, platforms and application. These networks can interact with each other, external environment and with other people. (Please refer Lecture 53@14:07)

QUESTION 9:

Which of the following design objectives are considered while using IoT devices for industrial applications?

a. Latency
b. Throughput
c. Scalability
d. All of these

Correct Answer: d. All of these

Detailed Solution: To use an IoT device for industrial applications, the design objectives that are considered includes energy, latency, throughput, scalability, topology. (Please refer

Lecture 53@22:55)

QUESTION 10:

Intelligent transport system is least likely to render which of the following connectivity?

a. Vehicle-to-vehicle connectivity
b. Vehicle-to-sensor connectivity
c. Vehicle-to-road infrastructure
d. Vehicle-to-home connectivity

Correct Answer: d. Vehicle-to-home connectivity

Detailed Solution: ITS provides Vehicle-to-vehicle connectivity, Vehicle-to-sensor connectivity, Vehicle-to-road infrastructure and Vehicle-to-internet connectivity.

(Please refer Lecture 54@4:19)

QUESTION 11:

Intelligent radio modem (IRM) can be connected to an IP serial device but not to non-IP serial device to extend the capabilities necessary for monitoring and communicating with other technologies.

a. True

b. False

Correct Answer: b. False

Detailed Solution: Intelligent radio modem (IRM) can be connected to IP and non IP serial devices to extend the capabilities necessary for monitoring and communicating with other technologies. (Please refer Lecture 54@21:14)

QUESTION 12:

Wi-Fi, RFID and other wireless technologies are utilized for preventing mining accidents.

a. True
b. False

Correct Answer: a. True

Detailed Solution: To prevent accidents inside the mines, RFID, WiFi and other wireless

technologies are used. (Please refer Lecture 54@7:09)

QUESTION 13:

Fill in the blank.
Approximately around _____ of the total data available currently in the world is unstructured.

a. 20 %
b. 40 %
c. 60 %
d. 80%

Correct Answer: d. 80%

Detailed Solution: Unstructured data accounts for 80% of the total data available today in the world. (Please refer Lecture 55@9:23)

QUESTION 14:

In flow of data, after data generation and acquisition comes the storage of the data.

a. True
b. False

Correct Answer: a. True

Detailed Solution: The steps involved in flow of data includes generation, acquisition, followed by storage and finally analysis. (Please refer Lecture 55@18:51)

QUESTION 15:

Match the correct characteristics of the big data present in statement II with statement I:

STATEMENT I	STATEMENT II
1.Uploading 70 hours video to YouTube every minute	a. Visualization
2. On average 130 million of tweets per day	b. Volume
3. Presentation of data in graphical format	c. Variety
4. Most of the data being unstructured or semi-structured	d. Velocity

a. 1-a, 2-c, 3-b, 4-d
b. 1-b, 2-a, 3-d, 4-c
c. 1-b, 2-d, 3-a, 4-c

d. 1-a, 2-d, 3-b, 4-c

Correct Answer: c. 1-b, 2-d, 3-a, 4-c

Detailed Solution: In big data, uploading 70 hours video to YouTube every minute represents volume, on average 130 million of tweets per day represents velocity, Presentation of data in graphical format represents visualization and most of the data being unstructured or semi-structured represents Variety (Please refer Lecture 55@19:14)

**********END**********

Introduction to Internet of Things

Assignment-Week 12

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15** **Total marks: 15 X 1= 15**

QUESTION 1:

Audio recordings collected through the microphone are subject to quantitative analysis.

a. True

b. False

Correct Answer: b. False

Detailed Solution: Audio recordings collected through microphone are subject to qualitative analysis. (Please refer Lecture 56@4:12)

QUESTION 2:

Fill in the blanks. A statistical model depicts how a set of _____ variable is related to another set of _____ variable.

a. Random, normal

b. Normal, random

c. Discrete, normal

d. Random, random

Correct Answer: d. Random, random

Detailed Solution: A statistical model depicts how a set of random variables is related to another set of random variables.

(Please refer Lecture 56@7:49)

QUESTION 3:

Select the statement/s that denotes the example of dispersion measure.

Statement I: Range, Variable and Standard deviation

Statement II: Range and average absolute deviation

Statement III: Variance, Standard deviation and average absolute deviation

a. Statement I

b. Statement II

c. Statements I, II, and III

d. None of these

Correct Answer: b. Statement II

Detailed Solution: The example of dispersion measure includes Range, Variance and Standard deviation and average absolute deviation (Please refer Lecture 56@13:36)

QUESTION 4:

The different components of AgriSens includes –

a. Sensor and actuator unit

b. Wireless communication unit

c. Power management unit

d. All of these

Correct Answer: d. All of these

Detailed Solution: The different components of AgriSens includes sensor and actuator unit, wireless communication unit, and power management unit. (Please refer Lecture 57@7:44)

QUESTION 5:

The two relevant sensors directly used in agricultures are

a. Soil moisture and proximity sensor

b. Soil moisture and water level sensor

c. ECG sensor an water level sensor

d. All of these

Correct Answer: b. Soil moisture and water level sensor

Detailed Solution: Soil moisture and water level sensors are the necessary sensors generally used in agriculture. (Please refer Lecture 57@13:11)

QUESTION 6:

The sensor node of AgriSens sensed the average soil moisture in –

a. Vegetative phase

b. Reproductive phase

c. Maturity phase

d. All of these

Correct Answer: d. All of these

Detailed Solution: The sensor node of AgriSens sensed the average soil moisture in vegetative phase, reproductive phase and maturity phase (Please refer Lecture 57@18:33)

QUESTION 7:

Select the correct order of the component layers present in the IoT healthcare.

a. Sensing layer, cloud platform layer, aggregated layer, processing layer

b. Sensing layer, aggregated layer, processing layer, cloud platform layer

c. Aggregated layer, sensing layer, processing layer, cloud platform layer

d. Sensing layer, processing layer, aggregated layer, cloud platform layer

Correct Answer: b. Sensing layer, aggregated layer, processing layer, cloud platform layer

Detailed Solution: The sensing layer senses data and transmit it o the aggregation layer where the data are aggregated. The aggregated layer further transfers the data to the processing layer the data are processed and final sent to the cloud platform. (Please refer Lecture 58@8:25)

QUESTION 8:

Which of the following communication protocol is used in AmbuSens?

a. IEEE 802.15.4

b. 6LoWPAN

c. IEEE 802.15.1

d. IEEE 802.15.2

Correct Answer: c. IEEE 802.15.1

Detailed Solution: In AmbuSens, the communication protocol used is Bluetooth i.e., IEEE 802.15.1 (Please refer Lecture 58@22:09)

QUESTION 9:

Wireless IoT driven solutions for remote healthcare facility provisioning brings healthcare to patients than bringing patients to healthcare.

a. True

b. False

Correct Answer: a. True

Detailed Solution: In IoT healthcare, wireless IoT driven solutions brings healthcare to patients than bringing patients to healthcare. (Please refer Lecture 58@11:40)

QUESTION 10:

Fill in the blanks. The physical activity tracking is a necessary component for _____.

a. Activity monitoring

b. Vehicle monitoring

c. Agriculture monitoring

d. All of these

Correct Answer: a. Activity monitoring

Detailed Solution: The physical activity tracking is a necessary component for activity monitoring (Please refer Lecture 59@6:39)

QUESTION 11:

Which of the following handheld devices are used for activity monitoring?

a. EEG and GPS

b. Accelerometer and EEG

c. Accelerometer and GPS

d. All of these

Correct Answer: c. Accelerometer and GPS

Detailed Solution: High end smartphones are likely to have accelerometer, compass, and gyroscope. (Please refer Lecture 59@15:29)

QUESTION 12:

Which of the following is the primary disadvantage of using camera-based activity tracking?

a. Accurate

b. Processing intensive

c. Expensive

d. None of these

Correct Answer: b. Processing intensive

Detailed Solution: Camera based activity tracking is very much process intensive. (Please refer Lecture 59@17:43)

QUESTION 13:

By performing continuous monitoring of a person's activity, it is not possible to observe his/her behaviour or to identify any repetitive pattern in his/her day-to-day activity.

a. True

b. False

Correct Answer: b. False

Detailed Solution: Continuous monitoring of activity results in daily observations of human behaviour and repetitive patterns in their activity. (Please refer Lecture 59@14:33)

QUESTION 14:

Which of the following are the inbuilt sensors that are present in high end smartphones?

a. ECG and EEG

b. Accelerometer, proximity sensor, EEG

c. Accelerometer, Compass, and Gyroscope

d. Pressure sensor NPK sensor

Correct Answer: c. Accelerometer, Compass, and Gyroscope

Detailed Solution: High end smartphones are likely to have accelerometer, compass, and gyroscope. (Please refer Lecture 60@2:28)

QUESTION 15:

Fill in the blanks. Processing the handheld activity device data with artificial intelligence can be used for _____.

a. Fall detection

b. Heart rate detection

c. Vehicle detection

d. All of these

Correct Answer: a. Fall detection

Detailed Solution: Processing the handheld activity device data with artificial intelligence can be used for detecting sudden fall of a person. (Please refer Lecture 60@11:56)

**********END**********